

Order Embioptera

Web-spinners

This relatively small order consists of 11 families and only about 400 species. Web-spinners are gregarious, and live in soil, leaf litter, and under bark. They make extensive protective tunnels, using the silk made by glands in their front tarsi. Metamorphosis is incomplete.

Family Clothodidae

Clothodids

Occurrence 15 spp. worldwide in tropical and subtropical regions; in soil, leaf litter, under bark and stones



Clothodids are typically elongated insects with short legs. They are 1/16–3/4 in (0.5–2 cm) long, and have small eyes, segmented antennae, and simple, biting mouthparts. As adults, males do not feed but use their

mandibles to hold onto the female during mating. Females and nymphs are saprophagous. Females cover their eggs with silk and detritus, and feed the nymphs with pre-chewed food.



ANTIPALURIA INTERMEDIA
Seen here inside its silken nest, this species is a native of Central America.

Order Zoraptera

Angel insects

These small, delicate, gregarious, termitelike insects make up a single family of 43 species. They are light straw-yellow to dark brown or blackish in color, with short, abdominal tails (cerci) and unspecialized mouthparts. Metamorphosis is incomplete.

Family Zorotypidae

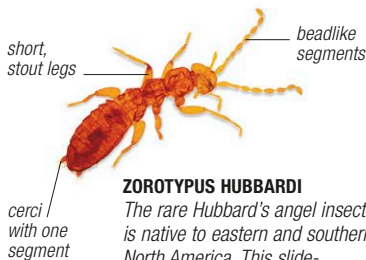
Angel insects

Occurrence 43 spp. worldwide in tropical and warm temperate regions except Australia; in rotting wood, sawdust



Angel insects are 1/16–1/8 in (2–3 mm) long, with a distinctive triangular head and a pair of antennae. The nymphs and adults eat fungal threads and small arthropods. Sexual behavior may be complex: males of some

species give the females mating gifts in the form of secretions from glands in their heads; in other species, the males fight for mates by kicking each other.



ZOROTYPUS HUBBARDI
The rare Hubbard's angel insect is native to eastern and southern North America. This slide-mounted specimen is stained red.

Order Psocoptera

Barklice and booklice

The order Psocoptera comprises 41 families and 5,600 species of soft-bodied, drably colored insects. The head is large, with bulging eyes, a bulbous forehead, and long, threadlike antennae. Metamorphosis is incomplete, typically with 5 nymphal stages.

Family Liposcelididae

Liposcelid booklice

Occurrence 250 spp. worldwide; in dry leaf litter, under bark, in nests, inside buildings, in food stores



These insects have flattened bodies, with swollen femora on the hindlegs. Most species are wingless. The length of the body ranges from 1/64–3/64 in (0.5–1.5 mm). Some species gnaw on

or chew stored flour, cereals, or paper. Eggs are laid in crevices in tree bark, leaf litter, and bird nests. The nymphs look like small adults.



LIPOSCELIS DECOLOR
Found in leaf litter and in a variety of stored, dry produce, this widespread species can become a pest.

Order Phthiraptera

Parasitic lice

Flattened and wingless, the 5,200 species (in 24 families) of parasitic lice are ectoparasites, living permanently on the bodies of birds and mammals without killing them. The mouthparts are used to bite skin, fur, or feathers, or to suck blood. Different lice are linked with specific hosts, and many are restricted to particular areas of the body. Parasitic lice develop by incomplete metamorphosis.

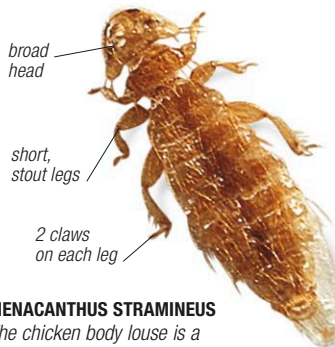
Family Menoponidae

Bird lice

Occurrence 1,153 spp. worldwide; on a variety of bird hosts



Bird lice are 1/32–1/4 in (1–6 mm) long, with a large and roughly triangular head, and biting mandibles. The abdomen is oval, and the legs are short and stout. Each leg has 2 claws. Adults and nymphs of most species feed on feather fragments but some also ingest blood and skin secretions. Eggs are glued in masses to the base of feathers. Some species, such as *Menopon gallinae*, or the shaft louse, can be serious poultry pests.



MENACANTHUS STRAMINEUS
The chicken body louse is a widespread species that lives on poultry, bringing about feather loss and infection.

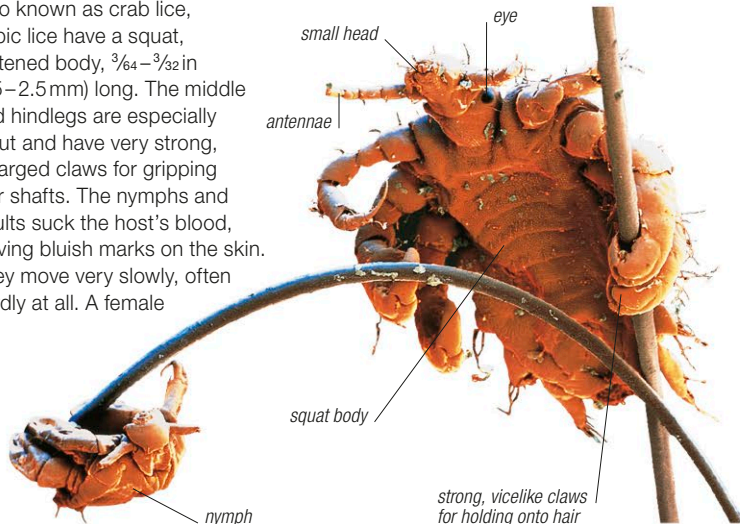
Family Pthiridae

Pubic lice

Occurrence 2 spp. worldwide; wherever their hosts (humans and gorillas) live



Also known as crab lice, pubic lice have a squat, flattened body, 3/64–3/32 in (1.5–2.5 mm) long. The middle and hindlegs are especially stout and have very strong, enlarged claws for gripping hair shafts. The nymphs and adults suck the host's blood, leaving bluish marks on the skin. They move very slowly, often hardly at all. A female



Family Pediculidae

Human lice

Occurrence 7 spp. worldwide; on humans, apes, monkeys



Human lice are small, pale, and elongated, measuring 1/16–1/4 in (2–6 mm) in length, with short, strongly clawed legs for gripping onto their hosts. They have a narrow head and a pear-shaped, flattened body. Adults and nymphs feed on blood. The human louse has 2 subspecies: the body louse, which glues its eggs to clothing, and the head louse, which glues its eggs (nits) to hair. Outbreaks of head lice are common among young schoolchildren. The body louse carries the organism that causes typhus.

PEDICULUS HUMANUS CAPITIS

A female human head louse can lay 9–10 eggs a day, fastening each one separately to a hair. Once cemented in position, the eggs are difficult to dislodge.



may lay 30 eggs in her lifetime, and uses a waterproof glue to stick them singly to pubic hair. It takes 4 weeks for the eggs to mature into adults.

PTHIRUS PUBIS

The pubic louse may be found in armpits and beards as well as the groin. Although unpleasant, it is not known to transmit disease.